

Redis Data Types — Study Notes

A practical reference for the five core Redis data types covered in the FastAPI + Redis lessons.

1. Mental Model

Type	Mental model	Typical use
String	One value	Cache, token, simple value
List	Ordered collection	Queues, recent items
Set	Unique collection	Unique users/tags/members
Hash	Field/value object	Object-like data
Sorted Set	Unique members + score	Leaderboards, rankings

2. Strings

A String stores a single value. It is commonly used for cached JSON, simple values, counters, and tokens.

```
SET product:1 "Mechanical Keyboard"
GET product:1
SET product:1 "Mechanical Keyboard" EX 60
TTL product:1
```

Key note: In the caching lesson, a Python dictionary was serialized with `json.dumps()` and stored as a Redis String.

3. Lists

A List is an ordered collection. LPUSH/RPUSH add items, while LPOP/RPOP remove items.

```
RPUSH jobs "send_email"
RPUSH jobs "generate_report" "resize_image"
LRANGE jobs 0 -1
LPOP jobs
LLEN jobs
```

Key note: Lists are useful for queue-like workflows. A common FIFO pattern is RPUSH by the producer and LPOP by the consumer.

4. Sets

A Set is an unordered collection of unique members. Adding the same member again does not create a duplicate.

```
SADD product:1:visitors alice
SADD product:1:visitors bob
SADD product:1:visitors alice
SMEMBERS product:1:visitors
```

```
SCARD product:1:visitors
SISMEMBER product:1:visitors alice
SREM product:1:visitors bob
```

Key note: Sets are ideal when uniqueness matters, such as unique product visitors, users who favorited an item, or tags.

5. Hashes

A Hash stores field/value pairs under one Redis key. It is a good fit for object-like data.

```
HSET product:1 name "Mechanical Keyboard" price 120 stock 25
HGET product:1 price
HGETALL product:1
HEXISTS product:1 stock
HDEL product:1 stock
```

Key note: With `decode_responses=True`, `redis-py` returns Redis string values as Python strings. Convert numeric fields with `int()` when needed.

6. Sorted Sets

A Sorted Set contains unique members, each associated with a score. Redis keeps members ordered by score.

```
ZADD leaderboard 100 alice
ZADD leaderboard 250 bob
ZADD leaderboard 180 charlie
ZRANGE leaderboard 0 -1 WITHSCORES
ZREVRANGE leaderboard 0 1 WITHSCORES
ZSCORE leaderboard alice
ZREVRANK leaderboard alice
```

Key note: Sorted Sets are useful for leaderboards, rankings, priority-style ordering, and other score-based collections. `ZREVRANK` uses a zero-based rank, so rank 0 means first place.

7. Command Cheat Sheet

String	SET	GET	DEL	TTL
List	LPUSH / RPUSH	LRANGE	LPOP / RPOP	LLEN
Set	SADD	SMEMBERS / SISMEMBER	SREM	SCARD
Hash	HSET	HGET / HGETALL	HDEL	HEXISTS
Sorted Set	ZADD	ZRANGE / ZREVRANGE	ZREM	ZSCORE / ZREVRANK

8. Choosing the Right Type

Need one value? → String

Need ordered items? → List

Need unique members? → Set

Need fields belonging to one object? → Hash

Need unique members ranked by score? → Sorted Set

9. Important Details to Remember

- Redis keys are case-sensitive: messages and MESSAGES are different keys.
- GET is for String values. Using GET on a List, Set, Hash, or Sorted Set produces a WRONGTYPE error.
- Lists preserve order; Sets do not guarantee an ordering you should rely on.
- Sets automatically remove the concept of duplicates: adding an existing member does not create another copy.
- Hashes are naturally suited to object-like field/value data.
- Sorted Set members are unique, but their scores can be updated.
- Sorted Set ranks are zero-based: rank 0 is first place.

10. FastAPI Examples

Set — unique visitors:

```
redis_client.sadd(f"product:{product_id}:visitors", user_id)
count = redis_client.scard(f"product:{product_id}:visitors")
```

Hash — product fields:

```
product = redis_client.hgetall(f"product:{product_id}")
redis_client.hset(f"product:{product_id}", "price", price)
```

Sorted Set — leaderboard:

```
redis_client.zadd("leaderboard", {"alice": 300})
redis_client.zrevrange("leaderboard", 0, 9, withscores=True)
```

Official References

Redis Data Types — <https://redis.io/docs/latest/develop/data-types/>

Redis Lists — <https://redis.io/docs/latest/develop/data-types/lists/>

Redis Sets — <https://redis.io/docs/latest/develop/data-types/sets/>

Redis Hashes — <https://redis.io/docs/latest/develop/data-types/hashtags/>

Redis Sorted Sets — <https://redis.io/docs/latest/develop/data-types/sorted-sets/>